Unit

19 Simple chemical cells

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Reactions in simple chemical cells
- Electrochemical series
- Salt bridge
- Daniell cell

19.1

Building simple chemical cells.

Reference Website

- Refer to the following websites for animations illustrating the chemical reactions occurring in a zinc-copper chemical cell:
  - http://group.chem.iastate.edu/Greenbowe/sections/projectfolder/animations/CuZncell.html (accessed July 2014)
  - http://www.mhhe.com/physsci/chemistry/essentialchemistry/flash/galvan5.swf (accessed July 2014)
- Refer to the following website for a computer simulation on constructing a chemical cell with different substances: http://group.chem.iastate.edu/Greenbowe/sections/projectfolder/flashfiles/electroChem/volticCell.html (accessed July 2014)

19.1 Reactions in simple chemical cells

In Unit 18, we learnt that a chemical cell is a device which uses chemical reactions to produce electricity. In this unit, we are going to find out how a simple chemical cell works.

A magnesium-copper chemical cell

From the reactivity series, we know that magnesium is more reactive than copper, so magnesium forms ions more readily than copper.

What changes might we observe in the magnesium-copper chemical cell shown in Fig. 19.1? Table 19.1 lists some observations and an interpretation of these observations.

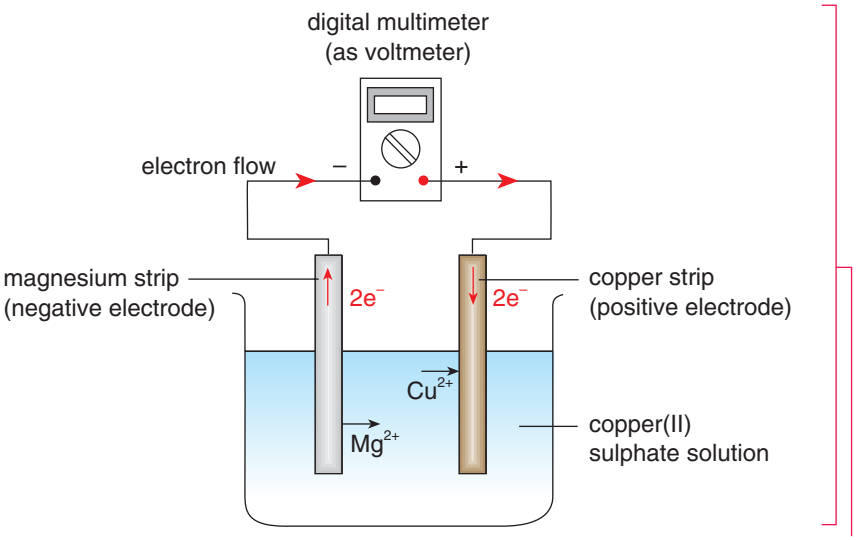
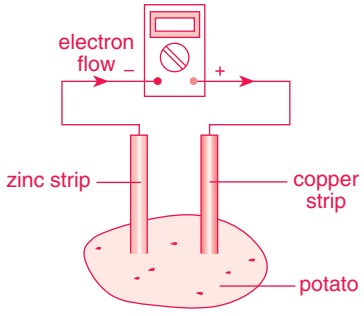


Fig. 19.1 A magnesium-copper chemical cell

DSE 2014 Paper 1B Q8(a)(i)

N1 Potato cells

Besides the lemon cell, the potato cell is another popular homemade cell.



A fresh potato should be used in the set-up as the fresh potato contains water so that the ions move more easily.

Two potato cells connected in series can be used to power a LCD clock. Notice that the positive electrode of one potato cell is connected to the negative electrode of the other potato cell (i.e. the cells are connected in series).

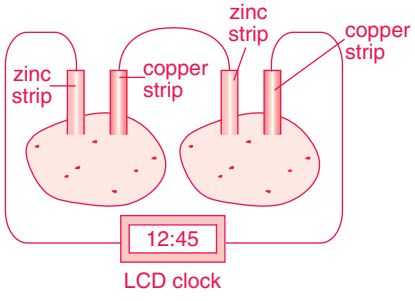


Table 19.1

Observations in a magnesium-copper chemical cell

Observation	Interpretation
• The mass of the magnesium strip decreases. DSE 2014 Paper 1B Q8(a)(i)	Magnesium atoms lose electrons and form magnesium ions. $Mg(s) \longrightarrow Mg^{2+}(aq) + 2e^-$
• The mass of the copper strip increases as copper is deposited on it.	Copper(II) ions in the solution gain electrons and form copper atoms. $Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$
• The digital multimeter gives a positive voltage.	The magnesium strip is the negative electrode while the copper strip is the positive electrode. Electrons flow from the magnesium strip to the copper strip in the external circuit. This flow of electrons forms an electric current.

We can obtain the overall equation for the reaction by combining the two **ionic half-equations**:

$$\begin{array}{l} Mg(s) \longrightarrow Mg^{2+}(aq) + \cancel{2e^-} \\ Cu^{2+}(aq) + \cancel{2e^-} \longrightarrow Cu(s) \\ \hline Mg(s) + Cu^{2+}(aq) \longrightarrow Mg^{2+}(aq) + Cu(s) \end{array}$$

In a magnesium-copper chemical cell, the magnesium and copper form a **metal couple**. We can represent a magnesium-copper metal couple by Mg/Cu.

A lemon cell

We can also build other simple chemical cells using the magnesium-copper metal couple. In the lemon cell shown in Fig. 19.2, copper(II) sulphate solution is replaced by lemon juice.

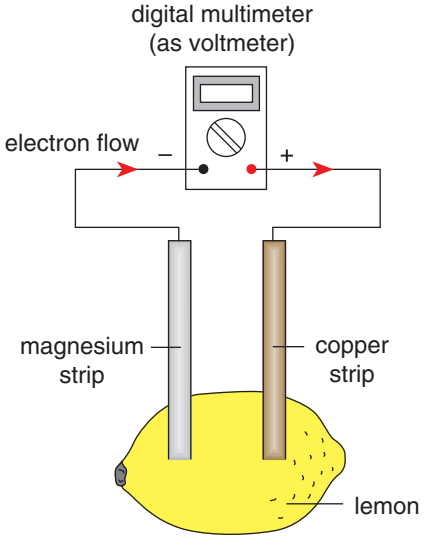


Fig. 19.2 A 'lemon cell' N1

ionic half-equation 離子半方程式 metal couple 金屬偶

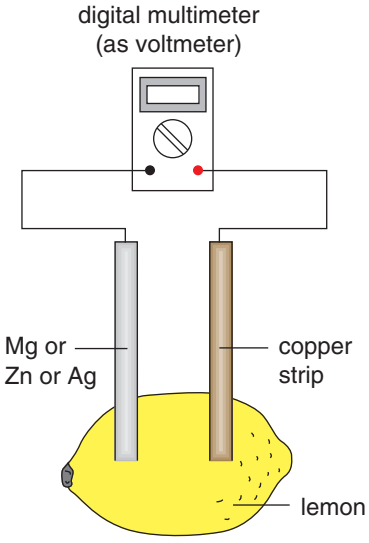


Fig. 19.3 A lemon cell made from a metal couple

19.2 Lemon cells made from different metal couples

The voltage of a cell depends on the metal couple used. We can compare the tendency of different metals to form ions by measuring the voltages of the lemon cells formed by coupling the metals with copper.

Fig. 19.3 shows the experimental set-up for measuring the voltages of lemon cells made from different metal couples (Mg/Cu, Zn/Cu and Ag/Cu). Table 19.2 lists the results obtained and the directions of electron flow in the external circuit of the cells.

Table 19.2

Voltages of lemon cells formed by couples different metal and the directions of electron flow

Metal couple	Voltage (V)	Direction of electron flow in the external circuit
Magnesium and copper (Mg/Cu)	+1.41	from magnesium to copper
Zinc and copper (Zn/Cu)	+0.79	from zinc to copper
Silver and copper (Ag/Cu)	–0.17	from copper to silver

In Table 19.2, the sign of the voltage indicates the polarity of the reference copper electrode. The voltages of the Mg/Cu and Zn/Cu cells have a positive sign. This suggests that copper is the positive electrode.

In these two cells, electrons flow from the magnesium and zinc electrodes to the copper electrodes in the external circuit. This is because magnesium and zinc form ions more readily than copper does.

The Mg/Cu cell gives a higher voltage than the Zn/Cu cell. Hence the difference between the tendencies for magnesium and copper to form ions is greater than that between zinc and copper. Magnesium forms ions more readily than zinc.

CE 2010 Paper 2 Q9

The voltage of the Ag/Cu cell has a negative sign. This suggests that copper is the negative electrode. In the Ag/Cu cell, electrons flow from the copper electrode to the silver electrode in the external circuit. This is because copper forms ions more readily than silver.

19.3 The electrochemical series of metals

Different metals have different tendencies to form ions. We can arrange metals in order of these tendencies. This is called the **electrochemical series** of metals (Table 19.3).

Table 19.3

Electrochemical series of metals

Metal ion				Metal	
$K^+(aq)$	+ e^-	$\rightleftharpoons$	$K(s)$		
$Ca^{2+}(aq)$	+ $2e^-$	$\rightleftharpoons$	$Ca(s)$	}	order is the reverse of that in the reactivity series
$Na^+(aq)$	+ e^-	$\rightleftharpoons$	$Na(s)$	}	
$Mg^{2+}(aq)$	+ $2e^-$	$\rightleftharpoons$	$Mg(s)$		
$Al^{3+}(aq)$	+ $3e^-$	$\rightleftharpoons$	$Al(s)$		
$Zn^{2+}(aq)$	+ $2e^-$	$\rightleftharpoons$	$Zn(s)$		
$Fe^{2+}(aq)$	+ $2e^-$	$\rightleftharpoons$	$Fe(s)$		
$Pb^{2+}(aq)$	+ $2e^-$	$\rightleftharpoons$	$Pb(s)$		
$2H^+(aq)$	+ $2e^-$	$\rightleftharpoons$	$H_2(g)$		
$Cu^{2+}(aq)$	+ $2e^-$	$\rightleftharpoons$	$Cu(s)$		
$Ag^+(aq)$	+ e^-	$\rightleftharpoons$	$Ag(s)$		
$Au^+(aq)$	+ e^-	$\rightleftharpoons$	$Au(s)$		

Decreasing tendency to form ions

A calcium atom loses electrons more readily in cell reactions than in reactions with air, water and dilute acids.

The electrochemical series of metals is actually drawn up using a hydrogen electrode as the reference. Hence hydrogen is included in the series even though it is not a metal.

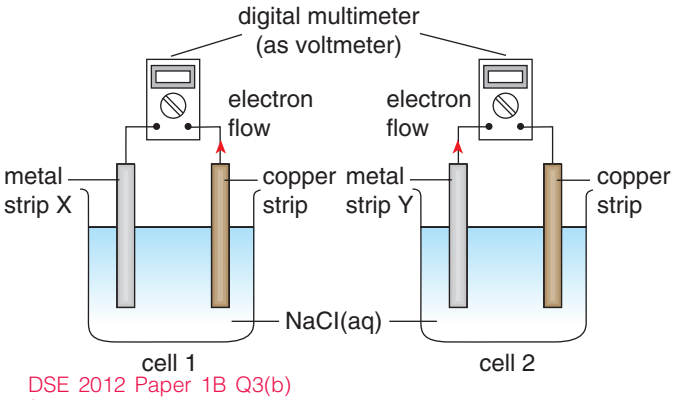
The order of metals in the electrochemical series is the same as that in the reactivity series (except calcium). This is because for a metal with a higher tendency to form ions, its atom will lose electrons more readily. Hence the metal is more reactive.

The voltage of a simple chemical cell is determined by the metal couple used. The farther apart the two metals are in the electrochemical series, the higher the voltage of the cell. The closer the two metals are in the electrochemical series, the lower the voltage of the cell.

electrochemical series 電化序

Example 19.1

Q The following diagram shows two simple chemical cells.



DSE 2012 Paper 1B Q3(b)

a) Arrange metal X, metal Y and copper in increasing order of tendency to form ions. Explain your answer.

b) Metal Y is a Group II metal.

i) Which metal strip in cell 2 is the negative electrode?

DSE 2012 Paper 1B Q3(c)(i)

ii) Write an ionic half-equation for the change that occurs at the negative electrode in cell 2.

A a) Order of tendency to form ions:

metal X < copper < metal Y

In cell 2, electrons flow from metal Y to copper. Thus, metal Y forms ions more readily than copper does.

In cell 1, electrons flow from copper to metal X. Thus, copper forms ions more readily than metal X does.

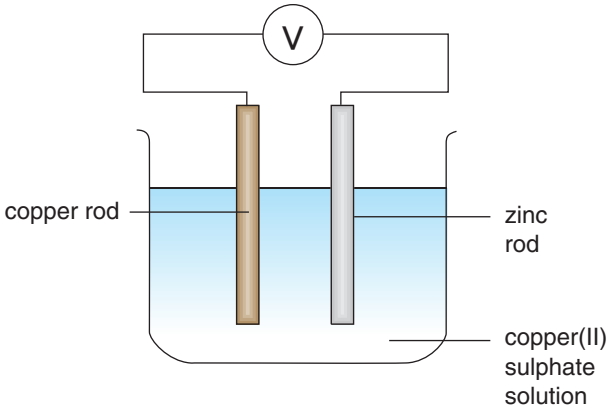
b) i) Metal strip Y is the negative electrode.

ii) $Y(s) \longrightarrow Y^{2+}(aq) + 2e^-$

Practice 19.1

P19.1 Refer to Teacher's Resources Book page 35.

A current flows through the external circuit of the chemical cell shown below.



a) Decide which electrode is the negative electrode of the cell.

b) What is the direction of electron flow in the external circuit?

c) Write an ionic half-equation for the process that occurs at

i) the copper rod; and

ii) the zinc rod.

d) The copper (II) sulphate solution is now replaced by a sugar solution.

What would happen to the voltage of the cell? Explain your answer.

19.4 Improving simple chemical cells

The Mg/Cu cell discussed in Section 19.1 does not work efficiently. There is a direct transfer of electrons from the magnesium electrode to the copper(II) ions in the solution. As a result, fewer and fewer electrons flow along the conducting wires. The voltage of the cell falls quite quickly and the electric current stops after a short time. Thus, the efficiency of this cell is low.

To solve the problem, we separate the cell into two **half-cells** as shown in Fig. 19.4. The magnesium and copper strips act as electrodes and are connected externally by a wire for electron flow. A **salt bridge** connects the two half-cells and completes the electric circuit. A salt bridge is simply a piece of filter paper soaked with an ionic salt solution such as saturated potassium nitrate solution.

Any ionic salt solution that does not react with the reactants of the cell can be used.

Animation
Chemical cell with a salt bridge

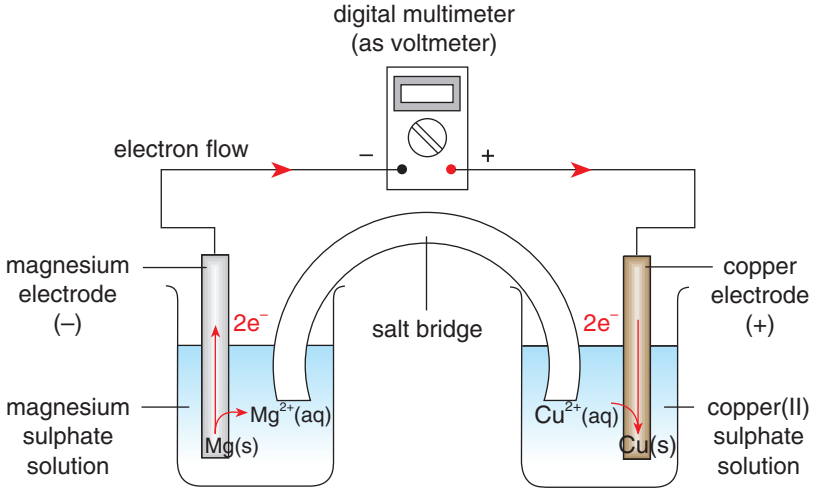


Fig. 19.4 A simple chemical cell separated into two half-cells

19.2
Determining the order of three metals in the electrochemical series.

19.5 The role of a salt bridge

Consider again the chemical cell shown in Fig. 19.4. In the magnesium half-cell, magnesium atoms on the surface of the magnesium electrode lose electrons and form $Mg^{2+}(aq)$ ions. This results in a build-up of positive charges.

In the copper half-cell, $Cu^{2+}(aq)$ ions gain electrons and form copper atoms. This results in a build-up of negative charges due to the $SO_4^{2-}(aq)$ ions that remain.

half-cell 半電池　　salt bridge 鹽橋

The salt bridge serves two important functions:

1 It completes the circuit by allowing ions to move from one half-cell to the other.

2 It provides ions that can move into the half-cells to prevent the build-up of charge in the solutions which would cause the reaction to stop (Fig. 19.5).

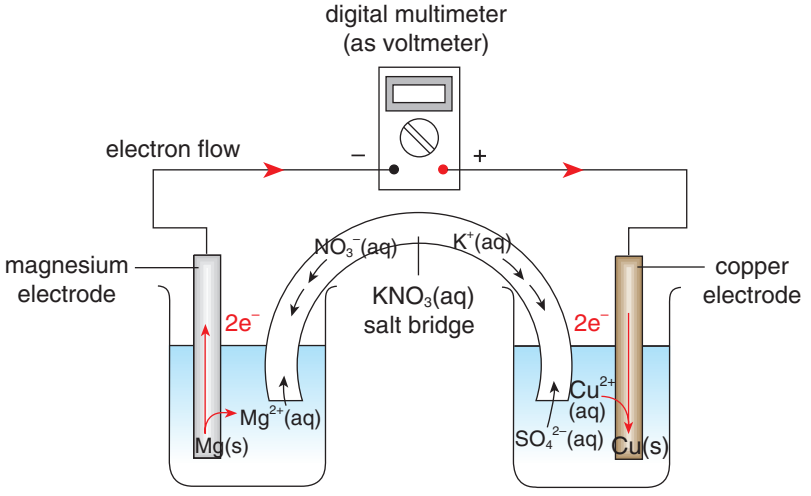


Fig. 19.5 A salt bridge providing ions to prevent the build-up of charge in the solutions

Notice that electrons move in the external circuit and the electrodes. Ions move in the salt bridge and the electrolytes.

As the cell operates, NO_3^-(aq) ions migrate from the salt bridge into the magnesium half-cell to offset any build-up of magnesium ions. Similarly, K^+(aq) ions migrate into the copper half-cell to offset any build-up of sulphate ions (Fig. 19.5).

Practice 19.2

P19.2 Refer to Teacher's Resources Book page 36.

CE 2011 Paper 2 Q7

X and Y are metals. Metal X reacts with Y^{2+}(aq) ions according to the following equation:

$X(s) + Y^{2+}(aq) \longrightarrow X^{2+}(aq) + Y(s)$

a) Which metal, X or Y, forms ions more readily?

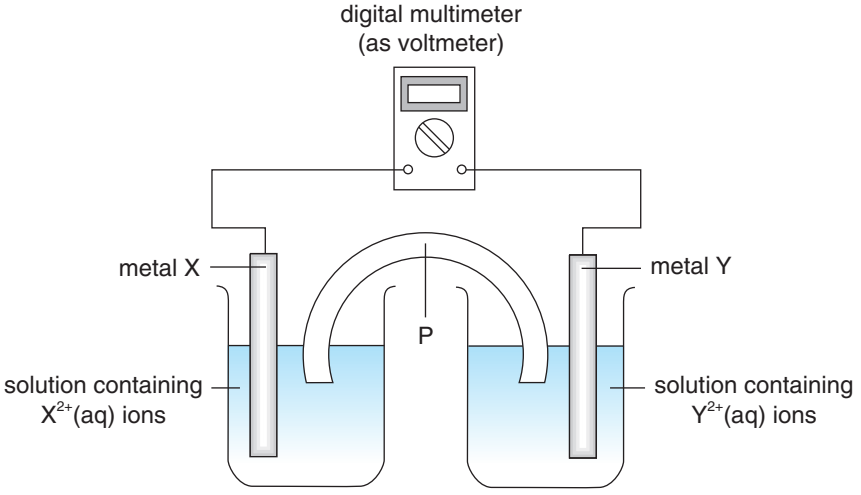
Continued on next page

b) Metals X and Y are used as electrodes in the chemical cell shown below.

i) What is P and what are its functions?

ii) Decide which electrode is the negative electrode of the cell. CE 2011 Paper 2 Q7

iii) What is the direction of electron flow in the external circuit?



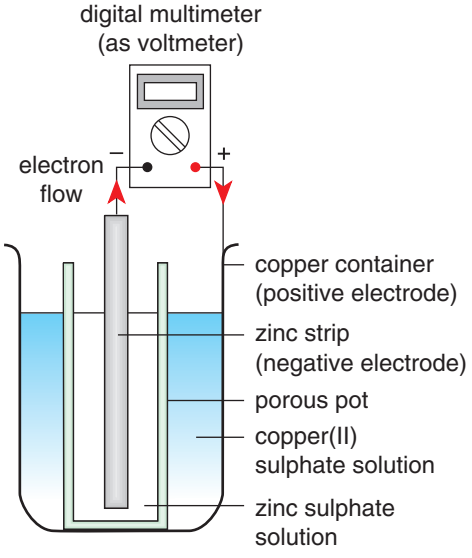


Fig. 19.6 A Daniell cell

A porous pot is different from a salt bridge. It does not provide ions to balance the charge in the solutions of the cell.

19.6 The Daniell cell

One of the earliest chemical cells invented was the **Daniell cell** (Fig. 19.6). It was developed in 1835.

In the Daniell cell, electrons flow from the zinc strip to the copper container in the external circuit. Zinc is the negative electrode. The copper container is the positive electrode.

At the zinc strip: $Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$

At the copper container: $Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$

The overall equation is:

$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$

The overall voltage of this cell is about 1.1 V.

The *porous pot* completes the circuit by allowing ions to move between the two solutions through its small holes.

Reference Website

Refer to the following website for an animation illustrating the chemical reactions that occur in a Daniell cell:
http://www.physics-chemistry-interactive-flash-animation.com/chemistry_interactive/daniell_cell.htm (accessed July 2014)

Daniell cell 丹聶爾電池　　porous pot 多孔瓷筒

Have you mastered?

Key terms

ionic half-equation 離子半方程式 21　　metal couple 金屬偶 21　　lemon cell 檸檬電池 21

electrochemical series 電化序 23　　half-cell 半電池 25　　salt bridge 鹽橋 25

Daniell cell 丹聶爾電池 27

Checklist

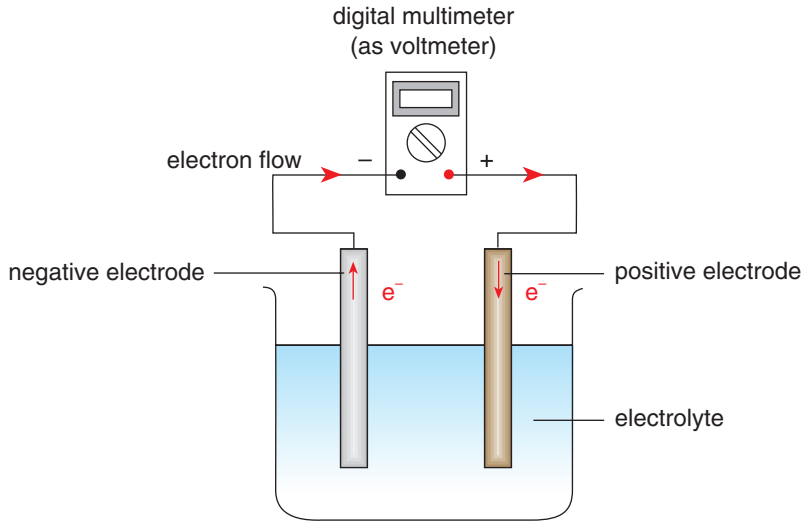
After studying this uint, you should be able to

- ☐ describe how to build simple chemical cells using metal electrodes and electrolytes;
- ☐ explain the differences in voltages produced in chemical cells when different metal couples are used as electrodes;
- ☐ write ionic half-equations and overall equations for simple chemical cells;
- ☐ predict the direction of election flow and chemical changes in simple chemical cells;
- ☐ explain the functions of a salt bridge / porous device;
- ☐ describe how to build simple chemical cells using metal-metal ion half-cells and salt bridges / porous devices.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 For a simple chemical cell made from two different metals, the metal which forms ions more readily is the negative electrode. The metal which forms ions less readily is the positive electrode.



Electrode	Electron flow	Electrode also called
Negative (metal which forms ions more readily)	from negative electrode to positive electrode in the external circuit	anode
Positive (metal which forms ions less readily)		cathode

2 Metals can be arranged in order of their tendencies to form ions. This is called the electrochemical series of metals.

3 The order of metals in the electrochemical series is the same as that in the reactivity series (except calcium).

4 To increase the efficiency of a simple chemical cell, the cell can be separated into two half-cells connected by a salt bridge.

5 A salt bridge serves two important functions:

a) It completes the circuit by allowing ions to move from one half-cell to the other.

b) It provides ions that can move into the half-cells to prevent the build-up of charge in the solutions which would cause the reaction to stop.

6 In the Daniell cell, electrons flow from the zinc strip to the copper container in the external circuit.

$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$

$Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$

Unit Exercise

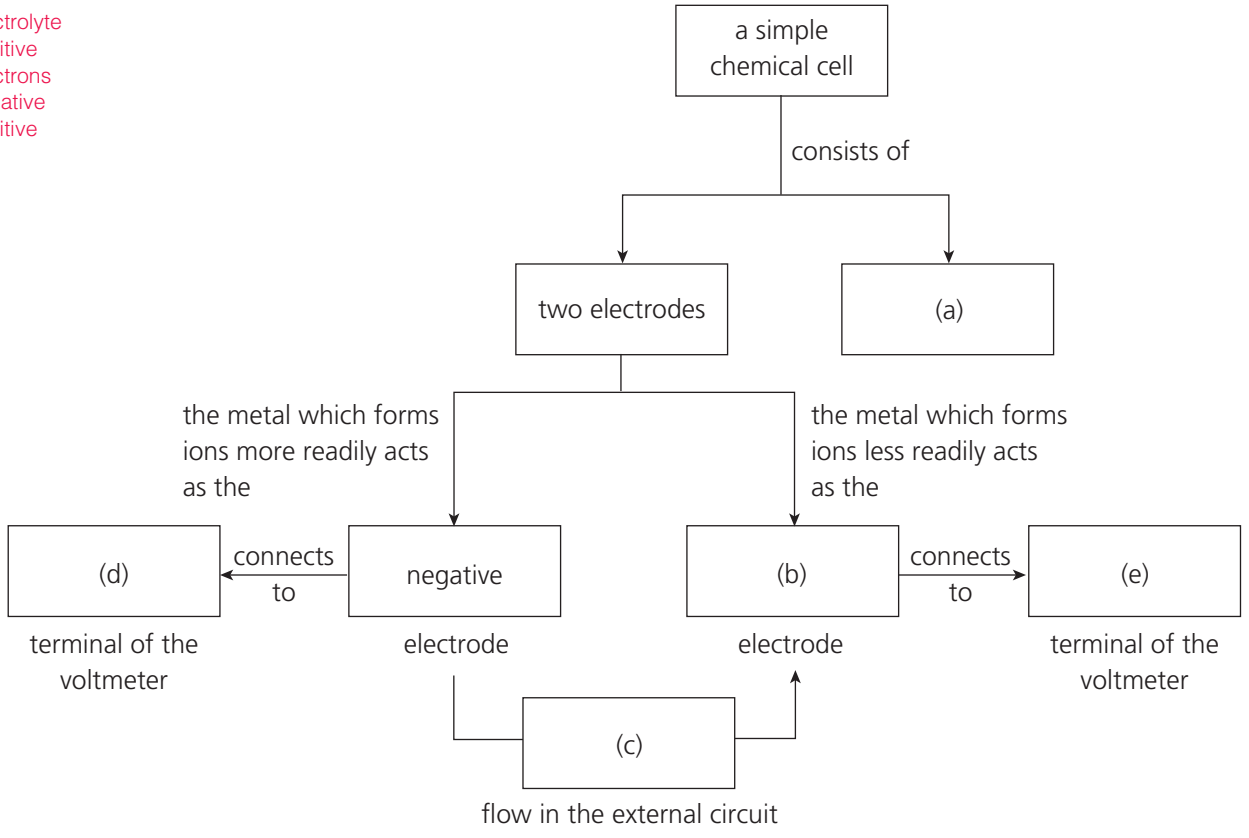
Refer to Teacher's Resources Book pages 36–39.

Note: The symbol ! indicates the level of difficulty of a question.

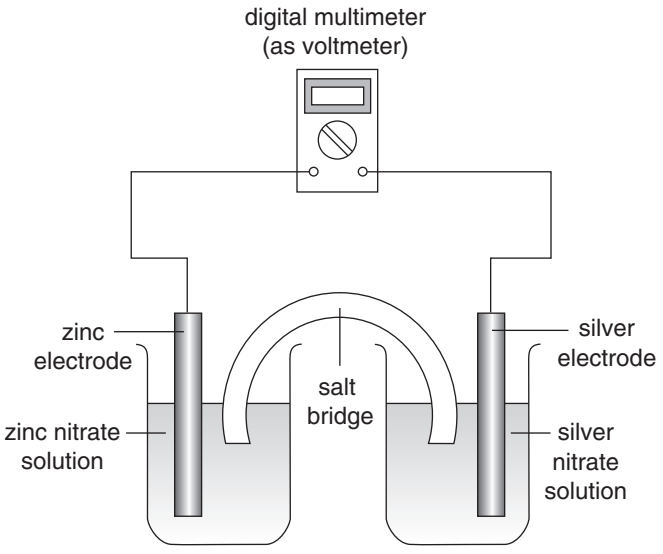
Part I Knowledge and understanding

1 Complete the following concept map.

a) electrolyte
b) positive
c) electrons
d) negative
e) positive



2 Consider the zinc-silver chemical cell shown below:



Complete the following statements about the above chemical cell.

a) Zinc atoms lose electrons to form zinc ions .

Ionic half-equation for the reaction that occurs at the zinc electrode:

$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$

b) Silver ions gain electrons to form silver .

Ionic half-equation for the reaction that occurs at the silver electrode:

$Ag^+(aq) + e^- \longrightarrow Ag(s)$

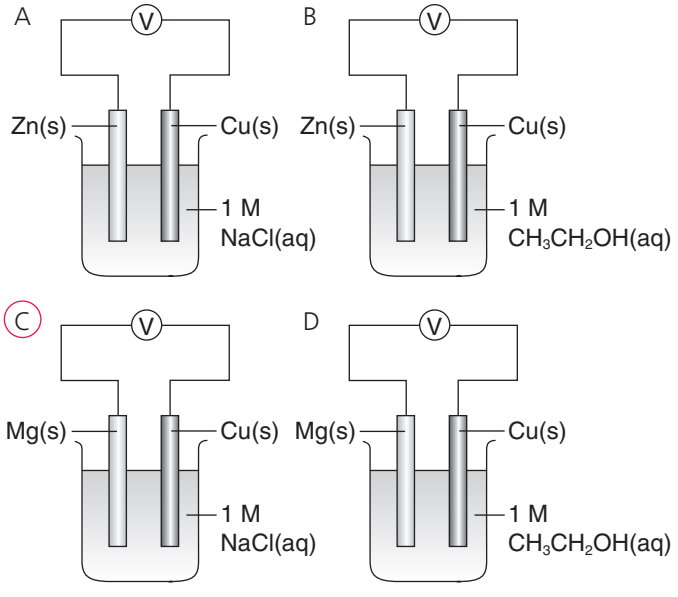
c) Electrons flow from the zinc electrode to the silver electrode in the external circuit.

d) The zinc electrode acts as the negative electrode while the silver electrode acts as the positive electrode.

Part II Multiple choice questions

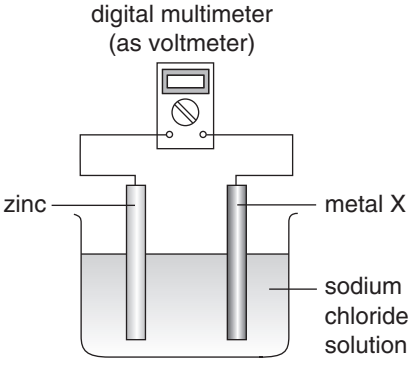
Refer to Teacher's Resources Book for explanations.

3 In which of the following set-ups would the voltmeter display the greatest magnitude of voltage reading?



(HKCEE, Paper 2, 2010, 9)

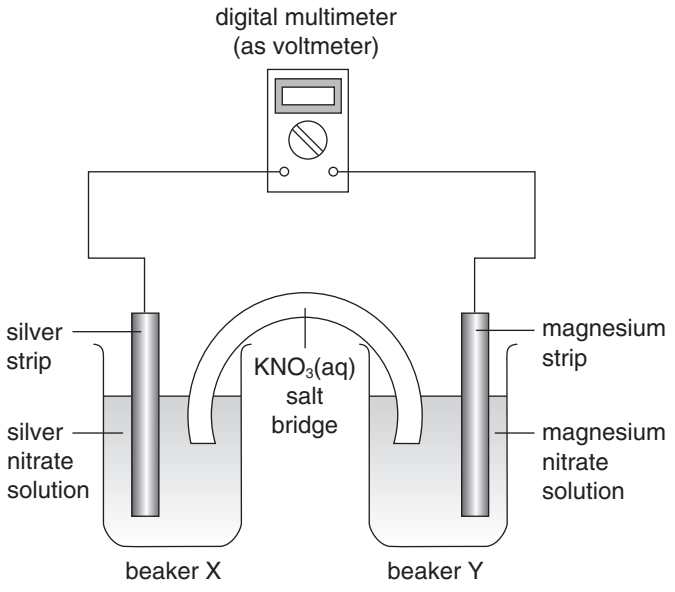
4 Consider the chemical cell shown below:



Which of the following combinations is correct?

	Metal X	Direction of electron flow in the external circuit
A	copper	from copper to zinc
B	iron	from iron to zinc
C	magnesium	from zinc to magnesium
(D)	silver	from zinc to silver

5 Consider the chemical cell shown below:



Which of the following statements about the chemical cell is correct?

A The silver strip acts as the negative electrode.
B The magnesium strip acts as the cathode.
C The concentration of silver ions in beaker X increases.
(D) Potassium ions migrate from the salt bridge into the silver half-cell.

6 Both D and E are metals. D reacts with ESO_4 solution according to the following equation:

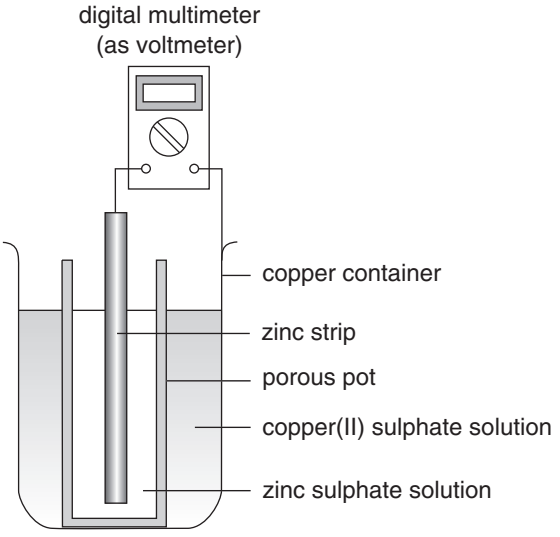
$$D(s) + ESO_4(aq) \longrightarrow DSO_4(aq) + E(s)$$

If D and E are used as the electrodes in a lemon cell, which of the following statements concerning the lemon cell during discharge is correct?

A Electrons flow from E to D in the external circuit.
(B) $D^{2+}(aq)$ ions are found in the lemon juice.
C E acts as the negative electrode.
D D acts as the cathode.

(HKCEE, Paper 2, 2011, 7)

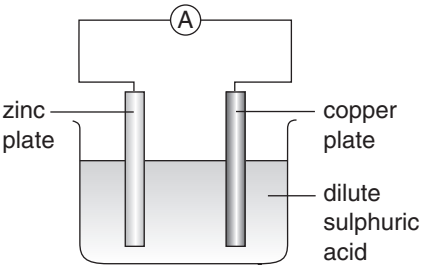
7 Consider the chemical cell shown below:



Which of the following statements about the chemical cell is correct?

A The zinc strip acts as the cathode.
(B) A current flows from the copper container to the zinc strip in the external circuit.
C The porous pot provides ions to prevent the build-up of charge in the two solutions.
D The blue colour of the copper(II) sulphate solution remains unchanged.

8 Consider the set-up shown below:

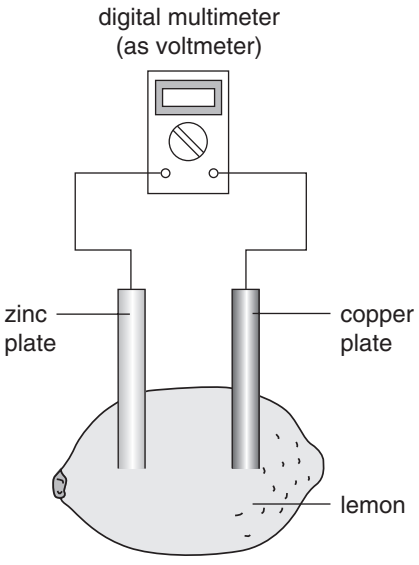


Which of the following statements about the set-up are correct?

(1) The zinc plate dissolves gradually.
(2) The hydrogen ions in the solution gain electrons to form hydrogen gas.
(3) The sulphuric acid turns blue gradually.

(A) (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

9 Consider the lemon cell shown below:

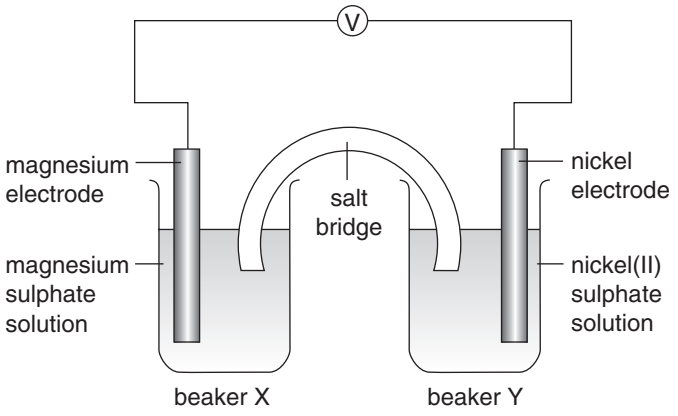


Which of the following statements about the lemon cell is / are correct?

(1) The lemon juice acts as the electrolyte.
(2) Electrons move from the zinc plate to the copper plate through the lemon juice.
(3) The voltage of the cell would increase if the zinc plate is replaced by a magnesium plate.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

10 Consider the chemical cell shown below:



It is known that magnesium is more reactive than nickel.

Which of the following statements about the chemical cell is / are correct?

(1) The concentration of magnesium ions in beaker X increases.
(2) The mass of the nickel electrode decreases gradually.
(3) The salt bridge allows electrons to flow from one half-cell to the other.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

Part III Structured questions

11 Consider the following ionic half-equations in the electrochemical series:

higher in electrochemical series

$Fe^{2+}(aq) + 2e^- \rightleftharpoons Fe(s)$

$Ni^{2+}(aq) + 2e^- \rightleftharpoons Ni(s)$

$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$

$Ag^{+}(aq) + e^- \rightleftharpoons Ag(s)$

lower in electrochemical series

a) Which metal on the list forms ions most readily?

b) Which metal on the list forms ions least readily?

c) Chemical cells can be constructed using the metal strips described in this question.

Suggest TWO metal strips you would choose to construct a chemical cell with the maximum output voltage. Explain your choice.

12 The relative positions of cadmium (Cd) and nickel (Ni) in the electrochemical series is shown below:

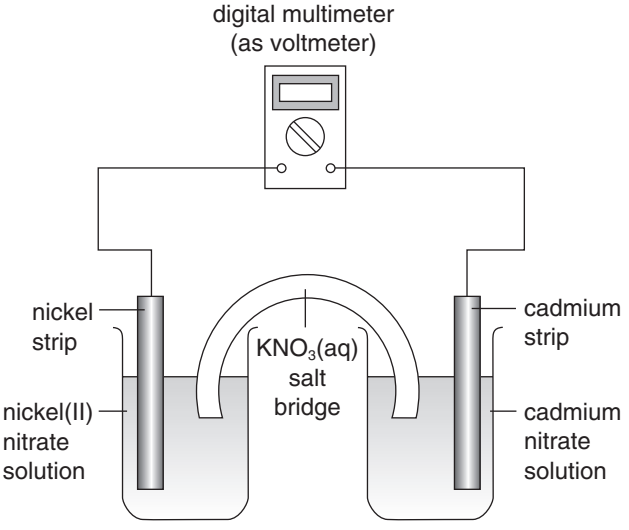
higher in electrochemical series

$Cd^{2+}(aq) + 2e^- \rightleftharpoons Cd(s)$

$Ni^{2+}(aq) + 2e^- \rightleftharpoons Ni(s)$

lower in electrochemical series

A chemical cell which consists of a cadmium strip and a nickel strip is shown below:



a) Which metal (cadmium or nickel) forms ions more readily?

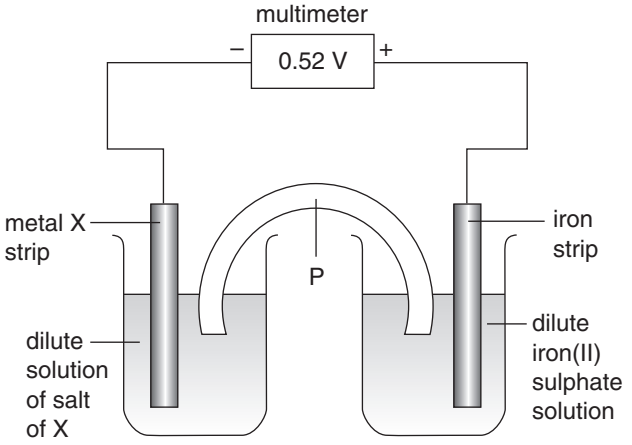
b) Write an ionic equation to show the overall cell reaction.

c) What is the direction of election flow in the external circuit?

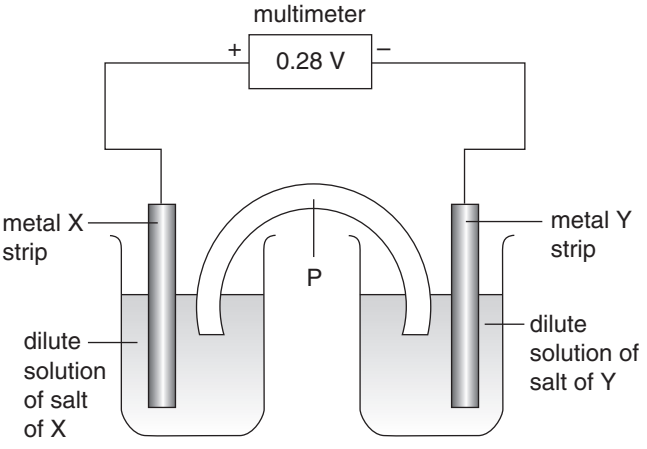
d) State and explain the direction of migration of $NO_3^-(aq)$ ions in the salt bridge.

13 X and Y are different metals. Consider the two chemical cells, A and B, shown below:

Chemical cell A:



Chemical cell B:



a) State TWO functions of P.

b) Arrange metal X, metal Y and iron in increasing order of tendency to form ions. Explain your answer.

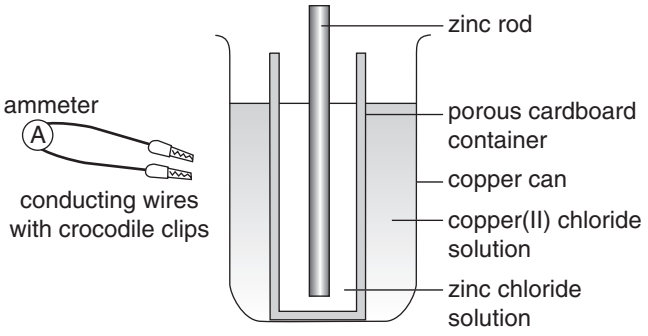
c) For chemical cell A,

i) write ionic half-equations for the reactions that occur at the iron strip and the metal X strip (X is a Group II metal);

ii) state and explain the colour change of the iron(II) sulphate solution after some time;

iii) what would happen to the multimeter reading if the iron strip and dilute iron(II) sulphate solution were replaced by a copper strip and dilute copper(II) sulphate solution respectively? Explain your answer.

14 A student wanted to show that chemicals can be used to produce an electric current using the following set-up.



When the crocodile clips were attached to certain parts of the set-up, the ammeter gave a reading.

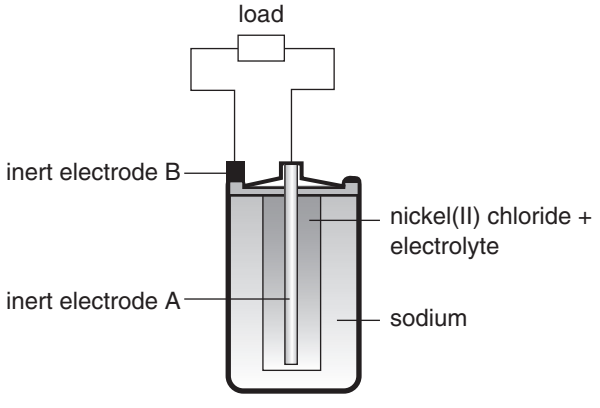
a) Suggest where the crocodile clips should have been attached.

b) Identify the direction of electron flow in the external circuit.

c) What would happen when the porous cardboard container was replaced by a glass beaker? Explain your answer.

d) What would happen to the reading on the ammeter if the zinc rod and zinc chloride solution were replaced by iron rod and iron(II) chloride solution respectively?

15 The diagram below shows a sodium-nickel(II) chloride cell connected to an external circuit. The cell operates at a high temperature of about 300 °C. The melting point of sodium is 98 °C while that of nickel(II) chloride is 1 001 °C.



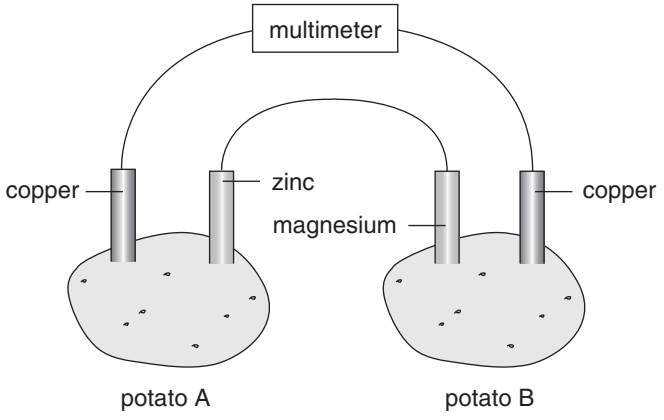
a) State and explain the direction of electron flow in the external circuit when the cell discharges.

b) Write an ionic half-equation for the process that occurs at the sodium electrode.

c) Suggest why it is necessary for the cell to operate at a high temperature.

d) Manufacturers suggest that sodium-nickel(II) chloride cells can be used to power cars. Give ONE advantage of this use.

16 The diagram below shows a set-up with metal strips inserted in fresh potatoes. The multimeter reading in the set-up is +0.75 V.



a) State, with explanation, the direction of electron flow across the connecting wire between zinc strip and magnesium strip.

b) i) Which metal strip in potato B is the anode? Why?

ii) Write the ionic half-equation for the change that occurred at the anode in potato B.

c) Which two metal strips should be interchanged in order to increase the multimeter reading?

d) Explain why fresh potatoes should be used in the set-up.

e) What will the multimeter reading be if the zinc strip in potato A is replaced by another magnesium strip, while the other three metal strips remain unchanged?

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